

Federated GAT-Twin: A Privacy-Preserving Cyber-Physical Framework for Resilient Smart Grid Anomaly Detection and Localization

Abstract

Modern smart grids are increasingly vulnerable to sophisticated False Data Injection Attacks (FDIAs) that manipulate sensor measurements while remaining undetected by traditional Bad Data Detection (BDD) mechanisms. To address this challenge, we propose *Federated GAT-Twin*, a privacy-preserving framework that integrates Graph Attention Networks (GATs), Digital Twins, and Federated Learning for real-time cyber-physical security monitoring. The proposed framework employs a dual-model architecture consisting of a graph-level Detection GNN for identifying grid-wide intrusions and a node-level Localization GNN for pinpointing compromised buses. In addition, we develop a physics-consistent FDIA generator to demonstrate how coordinated attacks can satisfy local power system constraints while bypassing conventional BDD methods. The framework is evaluated using both the MSU/ORNL 4-relay dataset and a simulated IEEE 118-bus transmission network. Results show that while traditional BDD remains ineffective against coordinated attacks, Federated GAT-Twin achieves 100% attack detection precision, 99.99% localization accuracy, and an average inference latency of 2.66 ms, well within real-time grid stability requirements. These findings demonstrate the effectiveness of Federated GAT-Twin as a scalable and practical defense for next-generation smart grids.

Keywords: Smart Grid Security, False Data Injection Attacks, Graph Attention Networks, Digital Twins, Federated Learning, Cyber-Physical Systems.